Battery Management System using 30kΩ Thermistor with Physical input from 60ºC-75ºC

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Abstract*—Effective Battery Management System is very important for safety and longevity of Electronic Vehicle (EV). Uncontrolled or unmonitored thermal performance can create accidental fire, degrading battery health and the performance of the battery and the vehicle. This Battery Management System is specifically made using a NTC 30kΩ Thermistor to monitor the temperature. When the temperature is equal or greater to 60 ºC, a warning indicator will light up, notifying the user, and when the temperature is equal or greater than 75 ºC the system will cut the power to avoid damages. A signal conditioning circuit is used to get an accurate reading of the temperature. Additionally, this is designed using a Virtual Instrumentation coding LabVIEW and myRIO board for the block diagram and front panel.

***Keywords*—BMS, NTC 30kΩ, LabVIEW, myRIO, EV, LM358**

1. Introduction

Temperature is one of if not the most important physical parameters measured in electronics system, essential for everything from environmental monitoring to system safety management.Hence, a temperature sensor is used to measure temperature in systems that are highly sensitive to temperature changes.A thermistor is a type of resistor that changes it resistance in response to temperature change. Thermistors are specifically designed to be highly sensitive to thermal change, thus making it easy for a microcontroller to detect. Thermistors are also very simple to operate and require minimal external components and are cost effective.For this project, we use NTC (Negative Temperature Coefficient) that means the resistance decreases as temperature rises.This high sensitivity to thermal changes is particularly valuable for safety-critical systems, as it allows for the immediate detection of overheating before damage occurs. Consequently, this component is essential for our application, where battery temperature is very crucial and it needs to be monitored and controlled. The safe zones for a Li-po battery's charging temperature is between 0°C to 45°C, while discharging is 20°C to 60°C. Using NTC Thermistor 30kΩ as our sensor with setting the temperatures range from 60 ºC to 75 ºC, to make specific actions. The first scenario, when the temperature is equal or greater to 60 ºC, a warning indicator will light up, notifying the user. But when the temperature is equal or greater than 75 ºC , myRIO will send a signal to the 5V relay to cut the power, this will help to protect the whole system from damages.

2. Design of electronic instrumentation

To construct a temperature control system for the Battery Management System, the system has a temperature operating range of 60°C to 75°C. The system's detector will only be built with a 30k NTC thermistor. The detector
The circuit is supported with a 5V supply.

This resistance variation is processed by the analog signal conditioning circuit, which includes a Wheatstone bridge configuration and an LM358 operational amplifier to produce a stable and scaled analog voltage. The conditioned analog signal is then fed into the myRIO, where it is converted into a digital signal through the analog-to-digital converter (ADC) and processed by the control algorithm implemented in LabVIEW. Finally, the digital output is used to control external devices such as a relay for battery thermal management.

2.1 Flow chart and block diagram

Figure 1. Block Diagram for Battery Management System

Figure 1: Block Diagram for Battery Management System illustrates the block diagram of the battery thermal management temperature control system, starting from the physical heat input to digital control. The battery temperature acts as the physical input, which is converted into a change in resistance by the NTC thermistor, serving as the transducer.

This resistance variation is processed by the analog signal conditioning circuit, which includes a Wheatstone bridge configuration and an LM358 operational amplifier to produce a stable and scaled analog voltage. The

conditioned analog signal is then fed into the myRIO, where it is converted into a digital signal through the analog-to-digital converter (ADC) and processed by the control algorithm implemented in LabVIEW. Finally, the digital output is used to control external devices such as a relay for battery thermal management.

Next, the electrical signal goes through Analog Conditioning. This stage includes a signal conditioner where the very small current signal from the NTC is amplified by an amplifier for easy processing and monitoring. The conditioned analog signal is then sent to the myRIO ADC. The myRIO ADC (Analog-to-Digital Converter) converts the analog electrical signal into a digital signal.

Finally, the digital signal is transmitted to the Digital Logic block. This block acts as the processor, which processes the digital signal. The processor is used to process the signal in various ways to facilitate signal transmission.

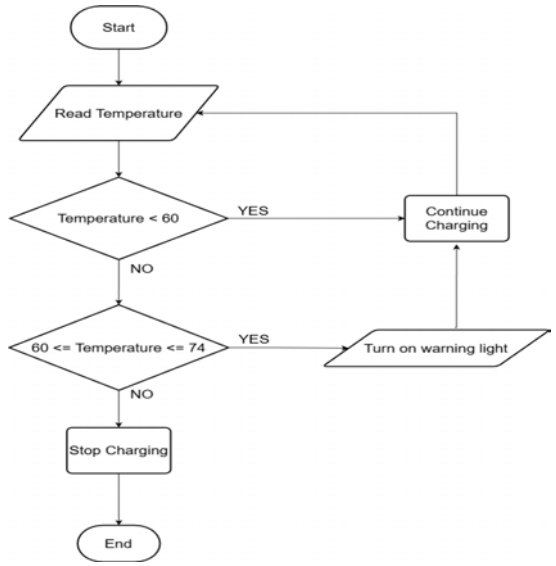

Figure 2. Flow Chart for Battery Management System

Upon power-up, the Battery Management System begins its monitoring cycle, using a 30kΩ NTC thermistor to sense the surrounding battery temperature and convert thermal variations into changes in resistance. This signal is processed by an analog conditioning circuit consisting of a Wheatstone bridge and an LM358 operational amplifier to produce a stable, amplified voltage that is then acquired by the myRIO ADC. The digital logic block within LabVIEW processes this data to display real-time status on the front panel, including temperature readings in Celsius, current resistance, and output voltage.

The control logic manages the battery's safety based on specific thermal thresholds. When the temperature is below 60°C, the system operates in a normal state, allowing the battery to continue charging. If the temperature rises into the warning range between 60°C and 74°C, the system activates a warning light to alert the user while maintaining operation. However, once the temperature exceeds 74°C, the protection mechanism is triggered; the digital logic commands the external relay to stop the charging process and disconnect the battery to prevent damage. This automated response ensures the system is well-suited for electric vehicle applications where maintaining thermal safety is critical.

2.2 Design of transducer

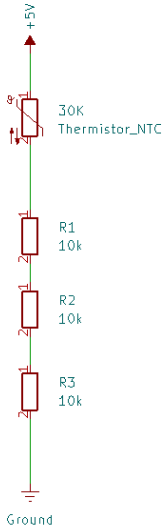

Figure 3. Transducer Circuit Diagram

$$V_{out} = V_{in} \times \frac{R_{ntc}}{R_{ntc} + R_{series}}$$

$$R_{ntc} = R_{series}\left(\frac{V_{in}}{V_{out}} - 1\right) \qquad (1)$$

From this derived voltage divider formula Equation (1),the resistance of the NTC thermistor can be calculated from the myRIO input analog signal,V_{out}. For this design, V_{in} or V_{cc} from myRIO is 5V the series resistors set to 30kΩ.

$$T_{kelvin} = \frac{1}{\frac{1}{T_0} + \frac{1}{\beta} ln\left(\frac{R_{ntc}}{R_0}\right)} \qquad (2)$$

Constants Used:

- T_0 (Reference Temp): 298.15 K or 25℃
- R_0 : 30kΩ at 25℃
- β : 4020 (MF58 Thermistor constant for 30kΩ.)

Finally, the temperature is converted to degrees Celsius. To illustrate how T_0 is derived, sample calculations for the thermistor at 25℃/30kΩ.

$$T_{celcius} = \frac{1}{\frac{1}{298.15} + \frac{1}{4020} ln\left(\frac{30k}{30k}\right)} - 273.15$$

$$T_{celcius} = 25\ ℃$$

For a more accurate reading, the signal out from the voltage through the thermistor is signaled into a differential amplifier.

2.3 Design of signal conditioning circuit

A signal conditioning circuit is used to get more accurate temperature reading. This is achieved by using Wheatstone bridge, powered by 5V to convert thermistor resistance into differential voltage. To establish this, by using R4 being 7.5kΩ and an NTC Thermistor 30kΩ. In the reference leg, a potentiometer is used to create a

reference signal A which then connects to a 10kΩ to pin 2 (-) on LM358. In between R4 and NTC is signal B which also is connected to a 10kΩ and goes to pin 3(+) on LM358.

By utilising a feedback resistance of 78kΩ and input of 10kΩ, the amplifier is configured with gain of 7.8. This small voltage boost converts into a distinct signal for the myRIO's analog input. Furthermore, the LM358 utilises the +15V rail to ensure the signal has sufficient headroom without saturation.

The NTC's resistance will decrease as the temperature rises. To achieve this specific calibration for the targeted temperature range of 60ºC to 75ºC. At these temperatures, the thermistor resistance was characterized as 7.45kΩ and 5.6kΩ, respectively. Using the voltage divider rule shown in Equation (3), these resistance values correspond to bridge output voltages of 2.146V (V_{60}) and 2.789 V (V_{75}).

$$V_{bridge} = V_{in}\left(\frac{R_{fixed}}{R_{NTC}+R_{fixed}}\right) \quad (3)$$

This yields a differential voltage span (ΔV) of 0.643 V. The gain required is calculated in Equation (4).

$$Gain = \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{5V}{0.643V} = 7.776 \approx 7.8 \quad (4)$$

The reference voltage for the differential amplifier is established at 2.146V to act as an offset null. This calibration ensures that at 60ºC (2.146 V_{input}) the output is effectively 0V, while at 75ºC (2.789 V_{input}), the output reaches approximately 5.015 V.

By using simulation we get this result. Figure 4. Temperature at 60ºC show that $V_{out} \approx 0V$ when $V_{bridge} = 2.146V$. Figure 5. Temperature at 75ºC shows that $V_{out} \approx 5V$ when $V_{bridge} = 2.789V$.

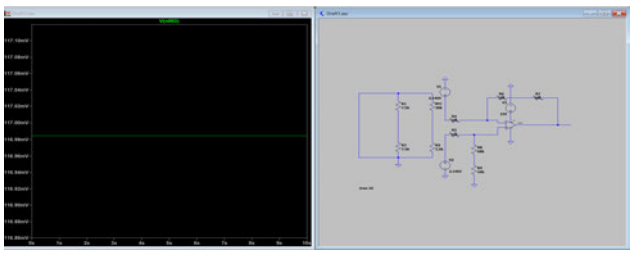

Figure 4. Temperature at 60ºC

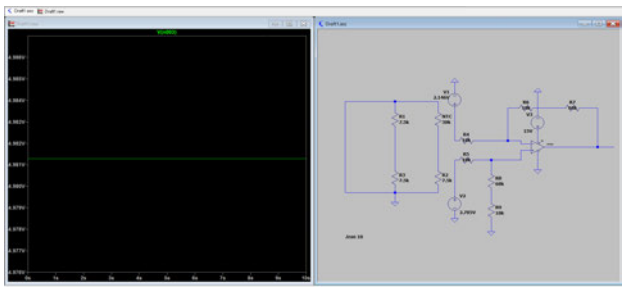

Figure 5. Temperature at 75ºC

2.4 List of components used

TABLE 1. List of the specifications components in Electronic Instrumentation Circuit

No	Components	Quantity
1	NI MyRIO Board	1
2	Breadboard	2
3	Wire	-
4	Operational Amplifier(LM358)	1
5	NTC Thermistor 30kΩ	1
6	10kΩ Resistor	4
7	7.5kΩ Resistor	3
8	68kΩ Resistor	2
9	Battery Li-Po 3.7V 800 mAH	1
10	Relay 5V	1
11	INA219 Current/ Voltage sensor	1
12	XL6009 (1.5-3.5)V Booster	1
13	USB-C Breakout	1
14	IP3212 3.7V Charger	1
15	Wago 221 Connector	-

2.5 Virtual instrument design using labview software and myrio

This project involves the design and implementation of virtual instruments by optimizing the synergy between LabVIEW and myRIO. LabVIEW provides a graphical development environment that interacts seamlessly with the myRIO's reconfigurable I/O capabilities, making it ideal for research and prototyping.The primary objective of Virtual Instrument Design is to to produce software managed hardware capable of handling real-time data acquisition, analysis and feedback control.This allows for suitable for a wide range of technical application.

2.5.1 Data Acquisition

Data acquisition refers to the process of measuring real-world physical phenomena. For this case, it is the temperature from the transducer or other external components to the myRIO board.

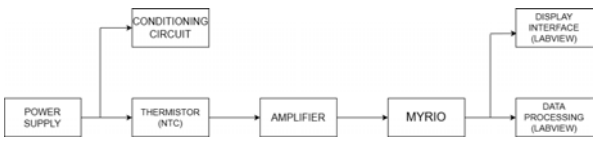

Figure 6 . Data Acquisition System Diagram

On the current system, the collected signals are from the differential amplifier output voltage (Vout), Thermistor

(NTC) output voltage (V2) and conditioning circuit output voltage (V1). The system uses a structured approach for acquiring analogue signals from various components:

1. Power Supply: Provides the necessary voltage and current to power theThermistor (NTC) and other components in the circuit.
2. Thermistor (NTC): used to sense the battery temperature by converting temperature changes into resistance variations that enable the system to monitor and protect the battery from overheating.
3. Conditioning Circuit:A Wheatstone bridge and LM358 amplifier convert NTC resistance into a scaled 0V to 5V signal to accurately monitor the 60°C to 75°C range.
4. Amplifier: Enhances the signal strength from the conditioning circuit to ensure it is suitable for further processing.
5. Microprocessor (MyRio): Receives the amplified analogue signal as input through its ADC (Analog-to-Digital Converter) channels. This device serves as the central DAQ unit, capturing the signal for processing in LabVIEW.

2.5.2 Data Processing

The data processing section of the system is implemented using myRIO and LabVIEW to acquire and process temperature sensor data, visualize battery temperature parameters, and control virtual and physical indicators as well as a relay based on predefined temperature thresholds.

Figure 7 . Data Acquisition System Diagram

Input Signal Processing

The signals for Temperature and Battery Voltage are obtained from the circuit and connected to the input channels of the MyRIO:

- Analog Channel A/AI0 for the Thermistor Voltage (V in).
- I2C Bus (Address 0x64) for the Battery Voltage reading.

Temperature Algorithm: The analog voltage is read and processed in real-time within a MathScript Node. The system calculates the true temperature by determining the NTC thermistor resistance (using a voltage divider with a 30kΩ resistor) and applying the Beta parameter equation ($\beta = 3950$) to convert resistance to degrees Celsius.

Battery Voltage Scaling: The I2C block retrieves two bytes of data (High and Low) from the battery sensor. These are joined into a single 16-bit integer and multiplied by a scaling factor of 0.004 to convert the digital value into a meaningful voltage output.

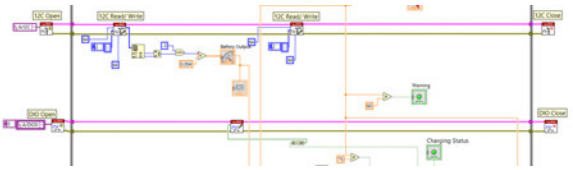

Figure 8 . Data Acquisition System Diagram

Control Logic & Alerts

Virtual LEDs (Boolean indicators) on the LabVIEW interface are utilized to indicate system status based on predefined safety thresholds:

- Warning Indicator: If the processed Temperature (℃)) exceeds 60, the "Warning" LED is activated.
- Charging Status: If the Battery Output exceeds 75 (voltage threshold), the "Charging Status" LED is activated.

The control logic continuously compares the processed sensor values against these constants to update the indicators in real-time.

2.5.3 Data display and presentation

Figure 9 shows acquired temperature and voltage. It used for monitoring and interaction

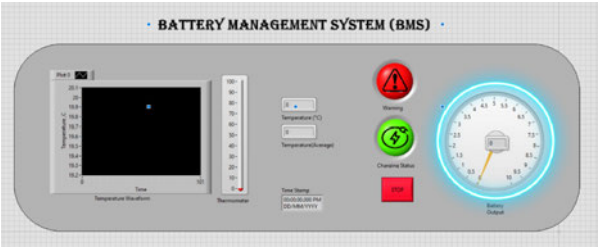

Figure 9 . Data Display Front Panel

Temperature Waveform (Graph)

- Visual: The black graphical plot on the left side of the dashboard.
- Function: It displays the history of temperature readings over time. This allows you to visualize trends, such as whether the battery is heating up or cooling down as the experiment progresses.
- Source: It receives the real-time TempC variable directly from the Formula Node.

Thermometer (Vertical Fill Slide)

- Visual: The vertical bar graph labeled "Thermometer" in the center.
- Function: It provides an analog-style visual representation of the current temperature relative to a scale (0 to 100). It acts as a quick visual check for the heat level.
- Source: Wired to the same TempC variable as the waveform.

Temperature (°C) (Numeric Indicator)

- Visual: The digital number box labeled "Temperature (°C)".
- Function: It displays the precise instantaneous value of the temperature in degrees Celsius (e.g., 24.5 °C) for the current loop iteration.

- Source: Wired to the TempC output from the Formula Node.

Temperature(Average) (Numeric Indicator)

- Visual: The digital number box labeled "Temperature(Average)".
- Function: Its intended purpose is to show the calculated average temperature (either moving average or cumulative) to smooth out sensor noise.
- Current State: In the specific block diagram provided (block diagram 15.1.png), this indicator is currently wired directly to the instantaneous TempC line.

Battery Output (Gauge)

- Visual: The large circular dial on the far right.
- Function: It displays the battery voltage (or current, depending on the sensor scaling) calculated from the I2C sensor data.
- Source: It gets its value from the I2C Read/Write block, where the raw data is processed using the formula (x - 3) x 0.004 to convert the digital signal into a readable physical unit.

Time Stamp (String Indicator)

- Visual: The box at the bottom center showing the time and date.
- Function: Displays the current system time for each data point recorded.
- Source: Generated by the "Get Date/Time In Seconds" function in the block diagram.

2.5.4 Data Logging

The data logging subsystem is designed to record critical Battery Management System (BMS) parameters for post-operational analysis. Unlike basic implementations that use Express VIs, this system utilizes LabVIEW's low-level File I/O functions to achieve precise control over file creation, data formatting, and storage efficiency.

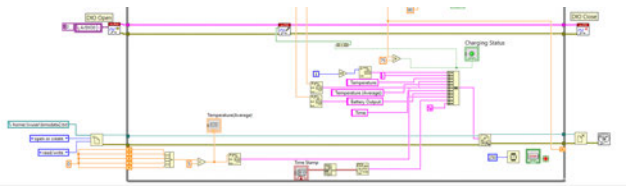

Figure 10 . Data Log Block Diagram

File Initialization and Management The logging process follows a structured "Open-Write-Close" architecture to ensure data integrity:

- Initialization: Before the main control loop begins, the Open/Create/Replace File function is executed. It is configured to "Open or Create" a file, ensuring that a log file is always available regardless of whether one already exists.
- File Path: The data is stored internally on the NI myRIO hardware at the path home:\lvuser\bmsdata2.txt. Storing the file locally on the device ensures the system remains portable and independent of a connected PC during operation.
- Termination: Upon pressing the "STOP" button, the loop terminates, and the Close File function executes immediately. This step is critical for saving the final data buffer and releasing the file resource to prevent corruption.

To generate a readable text log, the system uses the Format Into String function. This block aggregates multiple data types into a single, delimited string for each sample interval. The specific parameters logged include:

- Temperature: The instantaneous temperature reading derived from the thermistor logic.
- Temperature (Average): A running average of the temperature calculated to smooth out sensor noise.
- Battery Output: The voltage or current value acquired via the I2C communication block.
- Time: A timestamp generated by the "Get Date/Time in Seconds" function to correlate data points with specific events.

2.5.5 Actuator Control

The actuator control logic is designed to automatically interrupt the circuit if the battery temperature rises above the established safety limit. Using the temperature value calculated in the MathScript node, the system continuously compares the reading against the threshold. When this limit is exceeded, a boolean signal is sent through the myRIO's Digital Input/Output (DIO) channels to trigger the physical cut-off mechanism. To provide visual feedback, the software simultaneously activates the "Warning" indicator and deactivates the "Charging Status" light, ensuring the system clearly communicates that the charging process has been stopped due to a thermal fault. The design is as shown in Figure 11. Actuator Control Logic

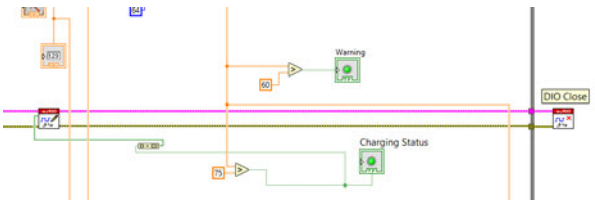

Figure 11. Actuator Control Logic

3. Result

The temperature monitoring system operates within a targeted temperature range of 60°C to 75°C. The system utilizes a Wheatstone bridge configuration fed into an LM358 differential amplifier with a calculated gain of 7.8. This signal conditioning setup amplifies the small differential voltage span into a full-scale 0-5V signal for the myRIO analog input.

TABLE 2.Results from design simulation compared with the actual measured value during testing

Temperature	Bridge Output, V_{bridge}	Theoretical Output ,V	Measured Output, V	Charging Status
60°C	2.146	0.00	0.08	Normal
75°C	2.789	5.015	4.98	Cutoff

As detailed in Table 2, the reference voltage for the differential amplifier was established at 2.146 V to act as an offset null.

At 60°C, the bridge output (2.146V) matches the reference voltage, resulting in a theoretical output of 0 V. The measured value of 0.08 V indicates a negligible offset due to real-world component tolerance.

At 75°C, the thermistor resistance drops to 5.6kΩ , raising the bridge voltage to 2.789 V. The amplifier multiplies this difference, resulting in a measured output of 4.98 V, which successfully triggers the safety cutoff logic(Figure 12 & 13) in the LabVIEW environment through the relay.

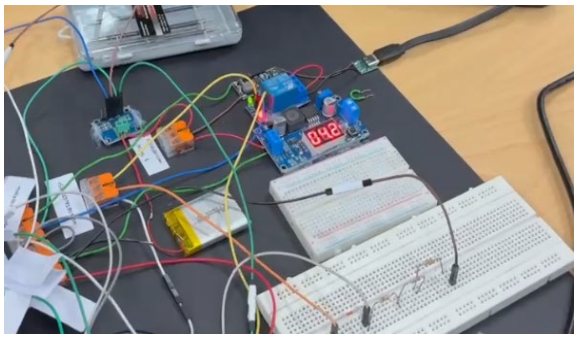

Figure 12. Relay status at temperature 60°C

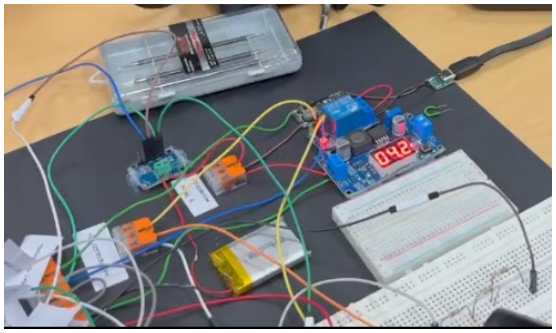

Figure 13. Relay status at temperature 75°C

4. Discussion

Our mini project aims to build a Battery Management System (BMS) that can check battery temperature and voltage accurately. A big challenge was getting the correct reading from the temperature sensor (NTC thermistor) because it does not give a simple linear output. To solve this, we used a MathScript Node in LabVIEW. By putting the formula directly into this block, we can convert the raw voltage from the sensor into a real temperature value. This makes sure the system displays the exact temperature in degrees Celsius, which is much more useful than just seeing a raw voltage number.

For the battery voltage, the challenge was that the I2C sensor gave us raw digital numbers instead of actual Volts. We fixed this by multiplying the raw data from the sensor (Address 0x64) by 0.004. This simple math turns the raw number into a readable voltage. We also wanted to stop the readings from jumping around too much (noise), so we added a "Mean" (Average) block. This smooths out the signal so the "Average Temperature" shown on the screen is stable and easy to read.

LabVIEW was very helpful for this project. The block coding made it easy to combine the analog temperature sensor and the digital voltage sensor into one system on the myRIO board. Because we could mix standard blocks with the text-based MathScript code, we were able to quickly process the data and show it on the screen in a way that is easy for humans to understand, like using gauges and warning lights.

5. Conclusion

In this project, myRIO is used as the main processing unit for an automatic temperature-based fan control system in a battery management application. A 30kΩ NTC thermistor is used to sense temperature, where its resistance decreases as temperature increases. The system is designed to operate within a temperature range of 60 °C to 75 °C, ensuring effective thermal monitoring and protection. Overall, the project demonstrates a reliable temperature sensing and control approach using MYRIO and proper signal conditioning.

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References

[1] "Product Documentation - NI." https://www.ni.com/docs/en-US/bundle/myrio-1900-getting-started/resource/376047d.pdf?srsltid=AfmBOorU0_l6_vh-zanHSf6iW2YhvjqzZ0wBhARiv4STIj2_hl8v4hor

[2] "BB1660TM, BB1660TTM Large BreadBoards," 660 tie point solderless plug-in breadboards provide a quick way to build and test circuits for experimentation or when learning electronics. [Online]. Available: https://components101.com/sites/default/files/component_datasheet/Breadboard%20Datasheet.pdf

[3] "LM358 data sheet, product information and support | TI.com." https://www.ti.com/product/LM358?utm_source=google&utm_medium=cpc&utm_campaign=asc-null-null-GPN_EN-cpc-pf-google-soas_en_cons&utm_content=LM358&ds_k=LM358+Datasheet&DCM=yes&gclsrc=aw.ds&gad_source=1&gad_campaignid=8754035891&gbraid=0AAAAAC068F2WO6HmOUm9LVITtF5H8D5Eq&gclid=CjwKCAiAvaLLBhBFEiwAYCNTfy

AkSRRYbmAbPxsuW9KjJvNWfCn6oo7JlVpz6o9WiGPKeOUdIWDXwxoCBX4QAvD_BwE

[4] AMWEI Thermistor Sensor, NTC 30K ohm Beta 3950K Radial Lead Epoxy Coated - AMWEI Thermistor Sensor. 2023. [Online]. Available: https://amwei.com/ntc-30k-ohm-beta-3950k-radial-lead-epoxy-coated/

[5] "LP284362 3.7V 800mAh Datasheet." [Online]. Available: https://www.li-polymer-battery.com/wp-content/uploads/2021/03/LP284362-3.7V-800mAh-Datasheet.pdf

[6] "SONGLE RELAY." [Online]. Available: https://components101.com/sites/default/files/component_datasheet/5V%20Relay%20Datasheet.pdf

[7] "INA219 data sheet, product information and support | TI.com." https://www.ti.com/product/INA219?utm_source=google&utm_medium=cpc&utm_campaign=asc-null-null-GPN_EN-cpc-pf-google-soas_en_cons&utm_content=INA219&ds_k=INA219+Datasheet&DCM=yes&gclsrc=aw.ds&gad_source=1&gad_campaignid=8754035891&gbraid=0AAAAAC068F2WO6HmOUm9LVITtF5H8D5Eq&gclid=CjwKCAiAvaLLBhBFEiwAYCNTf7SB3F92OAg3n6KeTQQFvSDzbA__3JgJ5QNaIp0dluAXtfnCMCNxgxoCRLYQAvD_BwE

[8] alldatasheet.com, "XL6009 Datasheet(PDF)," XLSEMI. https://www.alldatasheet.com/datasheet-pdf/pdf/1132228/XLSEMI/XL6009.html

[9] "USB-C-Breakout.sch." [Online]. Available: https://cdn.sparkfun.com/assets/6/1/7/9/8/USB-C-Breakout_Schematic.pdf

[10] "IP2312-4V35 Single cell Lithium Battery Charging IC," HUBTRONICS. https://hubtronics.in/ip2312?srsltid=AfmBOop57DnK4h8ngk02wN-zgQ8tIgtQVL6jJQiF3Yt_QAFRKu_fzj_V

[11] WAGO Kontakttechnik GmbH, "The 221 Series from WAGO – Flexible, Fast and Safe," WAGO. https://www.wago.com/global/electrical-interconnections/discover-installation-terminal-blocks-and-connectors/221